

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 50

Code of Conduct

I Shaik Ashraf (192324088) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly	Uses relevant	Uses vague or	No useful

		relevant cloud examples to strengthen the answer	examples with minor mismatch	partially relevant examples	examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Sol: Cloud Computing is the on demand delivering computing resources which is storage ,services, database And web service through the internet from provider to user .without the user owing or managing any physical infrastructure , instead of that we can use pays based on usage like renting the cloud platforms

Eg: Amazon AWS and Azure

Five characteristics

- 1.**On-demand Self Service:** A user can create and manage the resource without contacting the IT staff.
- 2.**Broad Network Access:** Cloud resources are accessible from anywhere via the internet, through any standard device.
- 3.**Resource Pooling:** we can share cloud resources through multiple users as needed.
- 4.**Rapid Elasticity:** Resources scale up or down automatically based on demand.
- 5.**Measured Service:** a user can pay as he uses the service.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Sol: Cloud computing didn't appear suddenly it's the result of several waves of technologies.

Types:

- 1.**Mainframe era:** Large, expensive mainframes were shared by multiple users via time-sharing terminals.
- 2.**Hardware Evolution:** This are minicomputers and PCs which are cheaper and smaller hardware.
- 3.**Networking and Internet:** TCP/IP and the internet allowed computers to communicate globally.
- 4.**Virtualization technology:** One physical machine could be split into several virtual machines.
- 5.**Cloud Platforms:** Cheap hardware + virtualization + internet + web-service delivery

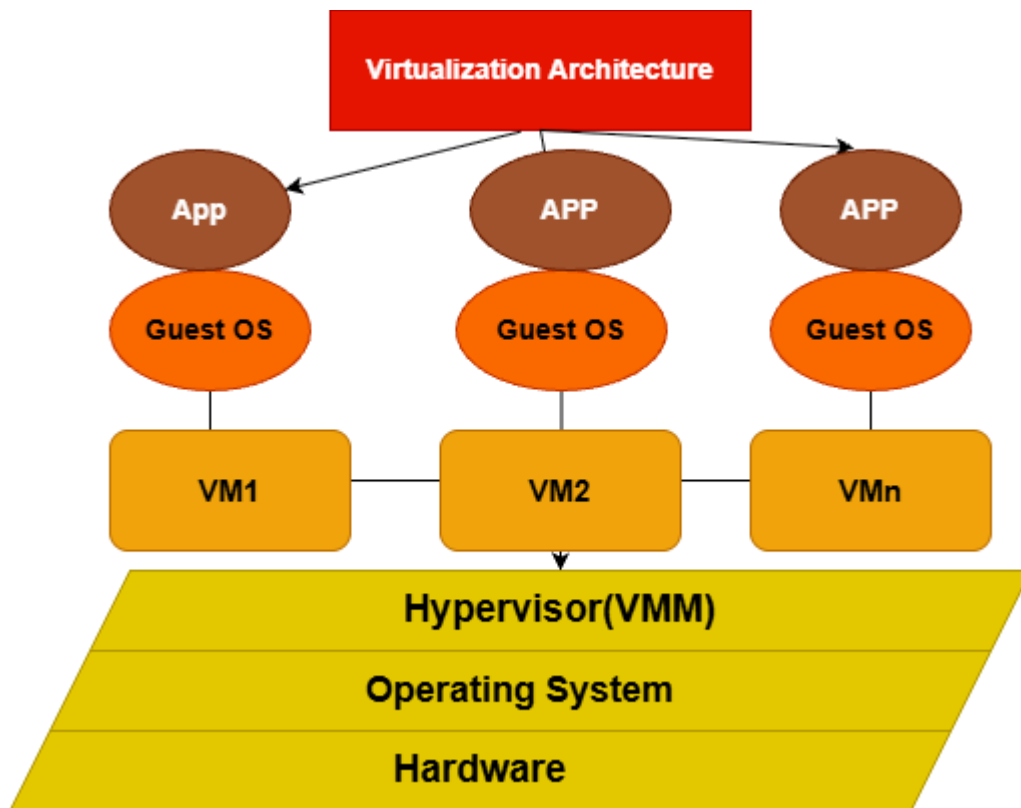
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Sol: Virtualization uses a hypervisor to split one physical server into multiple isolated virtual machines VMs, each running its own operating system and applications.

Features:

- Resource Pooling
- Elastic Scaling
- Cost Efficiency
- Isolation/Security

Eg: Hosted Architecture



4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.

Sol: Comparision between SOAP and REST

Role	SOAP	REST
Full Form	Simple Object Access Protocol	Representational State Transfer
Communication	Strict protocol using formally defined operations	Uses standard HTTP methods
Data Format	XML	Flexible- JSON ,XML and text
Ease of Use	Complex	Easier and Flexible
Security	Built in standards	Relies on HTTP
Examples	Banking System	Instagram and Twitter APIS

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5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Sol:

IaaS(Infrastructure as a Service): Provides virtual machines, storage, and networking the customer manages the OS
Eg: Amazon EC2

Paas(Platform as a Service): Provides a ready platform (OS, dev tools) so developers can only manage their application tools.
Eg: Google App

SaaS(Software as a Service): Delivers a complete, ready-to-use application over the internet; the provider manages everything.
Eg: Gamil and Microsoft 365

XaaS(Anything as a Service): Anything and Everything as a service which also provides Function as a Service , Identity as a Service and database as a Service.
Eg: Faas, IDaas